

Customer Segmentation — Business Insight & Action Plan

Methodology

RFM analysis (Recency, Frequency, Monetary) was applied to 2,500 households from the Dunnhumby Complete Journey dataset. Two clustering algorithms were evaluated:

Method	Clusters	Silhouette Score	Notes
KMeans	2	0.656	Clean but only Active / Churned
DBSCAN	3	0.758	Richer — discovers VIP outliers automatically

DBSCAN was selected as the final model due to its higher silhouette score and its ability to automatically detect the high-value VIP segment without specifying k.

Segment Summary

Segment	Recency	Frequency	Monetary	Size
Active Shoppers	22 days	103 baskets/yr	\$3,080 avg	2,431 (97%)
VIP	83 days	476 baskets/yr	\$10,304 avg	55 (2.2%)
Churned	544 days	4 baskets/yr	\$163 avg	14 (0.6%)

Action Plan per Segment

🏆 VIP (55 customers)

These 55 customers generate **3x the spend** of an average Active Shopper. Their top categories are Soft Drinks, Beef, and Domestic Wine — a notably different basket from regular customers.

Action: Launch a **VIP loyalty program** with exclusive early access, personalized offers on their top categories, and a dedicated account manager. Losing even one VIP customer costs the business ~\$10,000 in annual revenue.

🛒 Active Shoppers (2,431 customers)

The core customer base — shop every 22 days with consistent spend.

Action: Retention via personalized promotions on their top categories (Soft Drinks, Beef, Cheese). Focus on increasing basket size rather than visit frequency, which is already high.

⚠ **Churned (14 customers)**

Haven't shopped in ~544 days. Their historical categories (Beef, Beers/Ales, Chicken) differ from Active Shoppers, suggesting a distinct customer profile.

Action: Re-engagement campaign with coupons targeted at their last known purchase categories. Even recovering 50% of churned customers at their historical spend recovers ~**\$1,140 in annual revenue**.

Key Takeaway

Without segmentation, all 2,500 customers would receive the same promotion — wasting budget on Active Shoppers who already shop regularly and missing the opportunity to retain VIPs and re-engage churned customers with personalized offers. Segmentation enables the retailer to allocate marketing spend precisely where it has the highest return.